Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of an Inorganic Black Functional Nanomaterial Platform Technology for Enhancing Optical Performance of Semiconductor and Display Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and electronics		Electrical and electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Display	Display Components and Materials	Semiconductor Devices and Systems	Semiconductor Materials
Concept of Proposed Technology	• In semiconductor and display fabrication processes and optical systems, optical noise caused by stray light, internal reflection, and scattering has a direct impact on device performance and reliability. Therefore, black functional materials for suppressing such effects are an essential requirement. • Conventional black materials (typically carbon black) provide sufficient light-blocking performance; however, they exhibit limitations in simultaneously satisfying fine-process compatibility, process-condition stability, and long-term reliability. • For inorganic black functional nanomaterials, systematic control studies are required on particle surface states, formulation design, and dispersion behavior under different process conditions, as well as the resulting coating uniformity and variations in optical performance. • Accordingly, there is a strong need to move beyond simple material development toward the establishment of an **inorganic black functional nanomaterial (TiO_{2-x}) platform technology**, in which the oxygen bonding concentration (x) is precisely controlled to flexibly meet diverse industrial requirements.			

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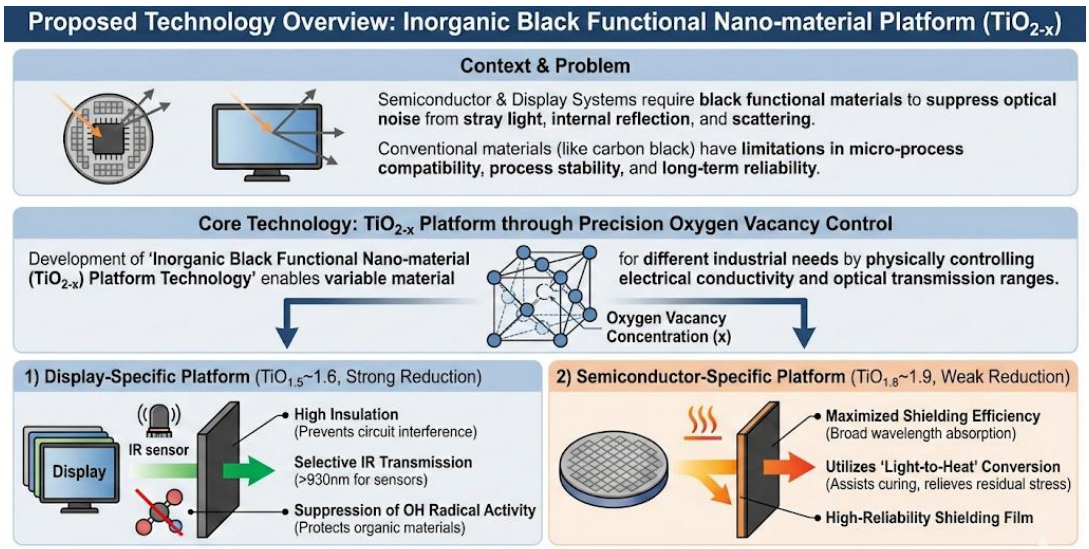
- In this project, by controlling the oxygen bonding concentration, the electrical conductivity and optical transmittance/absorption wavelength range of the material will be physically tuned, thereby enabling the implementation of differentiated core functionalities tailored to specific industrial applications.
- **1) Display-specific platform ($TiO_{1.5}$–$_{1.6}$, strongly reduced state):**
  This platform suppresses color change and degradation of material properties during dispersion in organic solvents, and prevents the decomposition of adjacent organic materials by blocking photocatalytic activity (•OH radical generation). In addition, it achieves effective absorption in the visible light region while simultaneously ensuring selective transmittance in specific wavelength ranges (≥930 nm, IR) to support next-generation sensor applications, as well as high electrical insulation properties to prevent circuit interference.
- **• 2) Semiconductor-specific platform ($TiO_{1.8}$–$_{1.9}$, weakly reduced state):**
  This platform maximizes light-blocking efficiency and effectively suppresses optical noise through broadband light absorption across a wide wavelength range, including the visible region. In particular, it utilizes the *light-to-heat* conversion mechanism that occurs when photo-excited electrons generated by visible-light absorption relax back to the substrate, thereby assisting densification of wafer coating layers and contributing to the formation of highly reliable light-shielding films through the alleviation of residual stress within the thin film.
- In summary, this project establishes a material platform technology applicable across semiconductor and display industries by systematically reflecting the analyzed dispersion behavior of inorganic black nanomaterials and their physicochemical property changes under different process conditions into material design.



Rationale for Joint R&D Project

- The dispersion stability and process-condition tolerance of black functional nanomaterials constitute a highly complex technological domain in which multiple factors—such as particle surface states, interfacial interactions, and formulation design—interact in a coupled manner.
- Dispersion collapse, structural evolution, and degradation of optical properties induced by process conditions (temperature, time, and atmosphere) are difficult to attribute to a single cause through short-term experiments or simple performance evaluations alone; therefore, advanced and highly precise analytical capabilities are essential.
- In particular, to optimize industry-specific and contrasting performance requirements—such as high electrical insulation and photocatalytic suppression for display applications versus high electrical conductivity and light-to-heat conversion functionality for semiconductor applications—within a single material platform, precise control of oxygen

	bonding concentration (TiO_{2-x}) combined with supporting interfacial engineering technologies is indispensable. • **Fraunhofer IAP** specializes in the dispersion of inorganic materials and the synthesis of surface-modified metal oxide materials. In particular, it possesses advanced infrastructure for the optical and physicochemical analysis of black functional materials under display and optoelectronic processing environments. Furthermore, Fraunhofer IAP has well-established analytical capabilities and accumulated expertise to precisely screen and elucidate degradation mechanisms in organic–inorganic hybrid materials. • This international joint research aims to establish a platform technology that satisfies the performance requirements of global end users by combining the proprietary oxygen-bonding-controlled material synthesis and property-tuning capabilities of domestic SMEs with the formulation and process-based analytical expertise of overseas research institutions.
Key Words	Inorganic black nanomaterials Dispersion stability Formulation design Optical noise suppression

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	• Identify particle surface and formulation-related factors that influence the dispersion stability of inorganic black functional nanomaterials (TiO_{2-x}), and establish an industry-tailored material property platform based on controlled oxygen bonding concentration. • Systematically analyze dispersion collapse behavior, structural evolution, and degradation of optical properties as a function of process conditions (temperature, time, and atmosphere). • Design high-functionality ink formulations capable of ensuring dispersion stability and tolerance to process conditions, enabling material implementation and thin-film formation. • Verify the applicability of the developed material platform to semiconductor and display manufacturing processes.			
R&D Target Specifications	• **Average reflectance in the visible region (400–700 nm):** ≤ 5% • **Dispersion stability:** Increase in average particle size after redispersion ≤ 20% (maintaining a stable dispersion state for ≥ 8 hours, evaluated by DLS) • **Solid content:** ≥ 10 wt% • **Optical property stability:** Change in reflectance before and after exposure to process conditions ≤ 10% • **Thickness uniformity:** within ±10% • **Optical density:** ≥ 2.5 at a film thickness of 1 μm			
R&D Plan	**<u>Research Approach and Strategy</u>** • This project will be conducted by reflecting the actual industrial process flow, progressing sequentially from material property control → formulation design → thin-film formation → exposure to process conditions. • Beyond securing single-property performance, the effects of oxygen bonding concentration control and material property tuning on the interrelationships among dispersion behavior, optical properties, and functional performance (electrical and thermal) will be systematically verified in a stepwise manner. • Optimal material property criteria will be established within a single platform to simultaneously address industry-specific and contrasting performance requirements, such as wavelength-selective transmittance and high electrical insulation for display applications versus light-to-heat conversion and electrical conductivity for semiconductor applications. **<u>Key Research Tasks and Methodologies</u>** **1. Industry-tailored material property control and dispersion process design** • Precise control of oxygen bonding concentration ($TiO_{1.5-1.9}$) and surface chemical states of inorganic black functional nanomaterials. • Implementation of high-insulation and photocatalytic (OH radical) suppression characteristics through strongly reduced $TiO_{1.5-1.6}$ material properties for display applications. • Maximization of electron transport efficiency and light-to-heat conversion functionality through weakly reduced $TiO_{1.8-1.9}$ material properties for semiconductor applications. • Establishment of optimal dispersion conditions through analysis of particle surface charge and interfacial interactions before and after material property tuning. • <u>Optimization of oxygen bonding concentration (e.g., $TiO_{1.5}$, $TiO_{1.7}$, $TiO_{1.8}$).</u>			

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2. Ink formulation design and redispersion property evaluation

- Design of ink formulations suitable for industrial organic solvent environments, considering high solid content (≥10 wt%).
- Precise monitoring of color change and optical property degradation during dispersion in organic solvents.
- Evaluation of particle size recovery upon redispersion and long-term dispersion stability under storage and transportation conditions.

3. Thin-film formation process development and process-condition tolerance evaluation

- Optimization of precision thin-film formation processes using slot-die coating, inkjet printing, and related techniques, ensuring thickness uniformity within ±10%.
- Analysis of property changes induced by exposure to actual industrial process temperatures and time conditions following thin-film formation.
- Verification of wavelength-selective transmittance (IR ≥ 930 nm) for display thin films, as well as evaluation of residual stress relaxation efficiency in semiconductor thin films.

4. Process Compatibility and Platform Validation

- Evaluation of process compatibility based on quantitative indicators, including reflectance (≤ 5%) and optical density (OD ≥ 4.0 at a film thickness of 1 μm).
- Derivation of industry-specific material property design criteria through repeated experiments and establishment of the versatility of the platform technology.
- Final validation of product performance in alignment with the specification requirements of global end-user companies.

Expected Milestones and Timeline

- **MS 1:** Optimization of oxygen bonding concentration by industry sector and design of surface property tuning processes.
- **MS 2:** Optimization of high-functionality ink formulations and establishment of dispersion stability in industrial organic solvent environments.
- **MS 3:** Application of thin-film formation processes and verification of tolerance to high-temperature process conditions..
- **MS 4:** Evaluation of performance against global end-user requirements and establishment of a material property platform technology.

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Infographic Illustration: Research and Development Plan: Strategy, Methodology, and Milestones

R&D Strategy

Material Modification → **Formulation Design** → **Thin Film Formation** → **Process Condition Exposure**

- Step-by-step verification of correlations between Dispersion-Optical-Functional (Electrical/Thermal) properties, beyond single property acquisition, through oxygen vacancy control & material modification.
- Establishing Optimal Modification Criteria within a single platform to meet diverse industrial needs (Display: Wavelength-selective transmission & Insulation vs. Semiconductor: Light-to-Heat conversion & Conductivity).

Main Research Content & Methodology

1. Customized Material Modification & Dispersion Process Design

- Precise control of oxygen vacancy concentration (TiO1.5~1.9) and surface chemical state.
- Display-specific Strong Reduction (TiO1.5~1.6): High Insulation & OH Radical suppression.
- Semiconductor-specific Weak Reduction (TiO1.8~1.9): Enhanced electron mobility & Light-to-Heat conversion.
- Optimal dispersion condition setting via analysis of surface charge & interfacial interactions before/after modification.
- Optimization of oxygen vacancy concentration control (e.g., TiO1.5, TiO1.7, TiO1.8).

2. Ink Formulation & Redispersion Characteristics Evaluation

- Ink formulation design considering high solid content (≥10 wt%) and industrial organic solvents.
- Precise monitoring of color & optical property changes during dispersion in organic solvents.
- Evaluation of particle size recovery & long-term dispersion stability upon redispersion considering storage & transport conditions.

3. Thin Film Formation Process Development & Process Condition Response Evaluation

- Optimization of precision thin film formation using Slot-die, Inkjet, etc., ensuring thickness uniformity (±10%).
- Analysis of property changes after thin film formation upon exposure to actual industrial process temperatures & times.
- Verification of wavelength-selective transmission (IR >930nm) for Display films & residual stress removal efficiency for Semiconductor films.

4. Process Suitability & Platform Validation

- Quantitative process suitability evaluation based on indicators like Reflectance (≤ 5%), Optical Density (OD ≥ 4.0/μm).
- Derivation of industrial material modification design criteria & establishment of platform technology universality through repeated experiments.
- Final prototype performance validation linked to global demand company specifications.

Expected Milestones & Schedule

- **Stage 1:** Optimization of industrial oxygen vacancy concentration & surface modification process design.
- **Stage 2:** Optimization of high-functional ink formulation & securing dispersion stability in industrial solvents.
- **Stage 3:** Application of thin film formation process & verification of response to high-temperature process conditions.
- **Stage 4:** Final performance evaluation based on global demand requirements & establishment of material modification platform technology.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Analysis of surface chemical states and interfacial structures of inorganic black functional nanomaterials (including particle surface bonding states, bonding/surface-layer evolution, and interpretation of interfacial interactions). • Development of surface treatment strategies and design of interfacial control concepts to ensure dispersion stability (including dispersant–particle interactions and directionality of surface property tuning). • Analysis of mechanisms governing changes in black optical properties (reflectance) before and after surface treatment and dispersion processes. • Design of surface property tuning strategies and interfacial control concepts to meet industry-specific requirements for semiconductor and display applications. • Quantitative evaluation of photocatalytic activity suppression mechanisms (OH radical generation) for display materials and verification of light-to-heat conversion efficiency for semiconductor materials. • Elucidation of degradation mechanisms related to changes in optical properties (e.g., reflectance) before and after material property tuning, as well as thin-film structural stability as a function of thermal treatment temperature. • Provision of technical feedback for material structure and formulation design based on analytical results. • Verification of applicability to various process-based thin-film formation techniques, including slot-die coating, spin coating, and inkjet printing, as well as evaluation of thickness uniformity.
Expected R&R of Korean SME Partner	• Synthesis of inorganic black functional nanomaterials with controlled TiO_{2-x} oxygen bonding concentration optimized for industry-specific requirements (e.g., high electrical insulation vs. electrical conductivity). • Implementation of platform-based material property tuning technologies to maximize optical performance and thermo-chemical stability of thin films. • Improvement of black optical characteristics and intrinsic thermo-chemical stability of the materials. • Realization of material property tuning and dispersion processes applicable to actual industrial environments by reflecting the surface and interfacial design concepts developed by Fraunhofer IAP. • Development of high-functionality ink formulations for thin-film formation with high solid content (≥ 10 wt%) and establishment of redispersion stability after storage. • Securing dispersion stability under practical use conditions during redispersion after storage.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes (Technical Achievements)** • Securing a platform technology for inorganic black functional materials that enables flexible realization of industry-specific performance requirements (high electrical insulation vs. electrical conductivity) through precise control of oxygen bonding concentration (x). • Establishment of design principles and a validated database for suppressing photocatalytic activity (OH radical generation) in display materials and achieving sensor-compatible wavelength selectivity (IR transmittance ≥ 930 nm). • Development of high-reliability light-shielding thin films for semiconductor applications by eliminating residual internal stress through light-to-heat conversion mechanisms. • Elucidation of optical and dispersion stability mechanisms under redispersion conditions and post–thin-film formation process conditions (150–200 °C). • Establishment of platform-based material design guidelines that reflect actual industrial process flows, from material property tuning to thin-film formation and subsequent process exposure. **Commercialization Potential (Industrial Impact)** • Securing direct product commercialization opportunities as black matrix (BM) materials for displays and as black coating materials for stray light suppression in optical systems. • Entry into the global semiconductor materials supply chain with high-performance black functional materials for stray light suppression and light-shielding thin-film formation. • Presentation of alternative and complementary solutions that overcome the limitations of conventional carbon-based black materials in terms of thermal stability and process compatibility. • Establishment of a customized business model that enables rapid response to diverse customer performance requirements through platform-based technology, solely by adjusting material property tuning conditions. • Expansion of the black functional material product portfolio of domestic SMEs and establishment of a foundation for mid- to long-term commercialization of advanced products. **Industrial and Economic Impact** • Enhancement of process reliability for essential black materials in semiconductor microfabrication and next-generation display industries, along with strengthened technological self-reliance in core materials. • Strengthening the technological competitiveness of domestic SMEs through the acquisition of proprietary material property tuning technologies in the high-value functional materials sector, and expansion of global collaboration networks. • Creation of spillover effects across related industries—including energy catalysts and high-sensitivity sensors—based on the scalability and extensibility of oxygen-bonding-controlled platform technologies. • Contribution to the technological paradigm shift of the domestic materials, components, and equipment (MCE) industry by presenting material design and property tuning platform models that explicitly incorporate process-condition considerations.

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6. References

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(3) Zhang, X., Kang, H., Jia, X., Wang, Q., Lu, S., Qin, W., & Wu, X., *Construction of broad-spectrum light absorption inorganic ultra-black coating with multi-scale light-absorbing structures based on hollow* $CuCr_2O_4$ *spheres*,

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization